

Supplementary Materials: Stress-Testing Data Minimization Across Review Budgets and Decision Stages in Educational Early Warning: Paired Evidence from OULAD and Two UCI Datasets

Aigul Shaikhanova ^{1,2}, Oleksandr Kuznetsov ^{3,4,*}, Gulmira Shangytbayeva ^{5,*}, Kamila Bakenova ^{1,6}, Kainizhamal Iklassova ⁷ and Dana Tulemisova ⁸

1. Supplementary Methods and Results

1.1. Evidence and provenance ledger

Table S1. Computational evidence layers retained during the correction.

Layer	Available evidence	Status	Treatment in this manuscript
v0.2.1	Official archive, complete code, saved aggregates	Exact replay on pinned bytes; all 64 workload rows reproduced	Establishes the baseline and forensic reference
C06/C08	Emergency aggregate summaries; two pages from C08	Models, predictions, full recipe, and full PDF unavailable; NOT_REVERIFIED	Not used as current evidence or numerical target
C11	Complete same-run code and outputs on derivative archive b344...	Superseded after missing-marker provenance error	Retained only to document the transition to C12
C12/C12E OULAD	Official archive f2ed..., complete same-run code, models, predictions, draws, and checks	Verified canonical OULAD execution; post-development exploratory	Sole source of current OULAD model and paired-statistical results
C13 UCI 697	Official archive e90e...; 54 configurations, models, OOF predictions, and paired draws	Fully verified on 2 September 2026; historical HGB workload replay exact	Separate external higher-education stage contrast; not pooled with OULAD
C13 UCI 320	Official outer archive 82ae...; current nested student.zip; 60 configurations	Fully verified on 2 September 2026; Mathematics and Portuguese linkage not unique	Subjects analyzed separately; negative and imprecise exclusion results retained
C05 XuetangX	Fully preserved v0.3.1 computational result	Official KDD Cup schema has no demographic fields; demographic derivative cannot be independently verified	Disqualified from demographic-exclusion evidence; preserved and not retrained

2. C14 Multi-Budget, Proxy, and Harmonized-Block Extension

2.1. Frozen C14 analysis status

C14 was specified after C12 and C13 results were available and is a post-development extension rather than preregistered confirmation. The protocol fixed review budgets of 5%, 10%, 20%, and 30%; targeted proxy outcomes; two operationally harmonized blocks; a same-run full-control replay tolerance of 10^{-12} ; 5,000 primary and 2,000 sensitivity bootstrap replicates; and Bonferroni-percentile families over every frozen stage and budget. XuetangX C05 was preserved without extraction or retraining and was excluded from demographic claims.

C14B used saved probabilities only. C14C fitted 44 fixed probes and retained 250,044 held-out predictions. C14D fitted 28 new HGB bundles and no new logistic or UCI 697 outcome models; its 12 distinct new full controls reproduced frozen probabilities with maximum absolute error 1.11×10^{-16} . C14E evaluated 151 contrasts in 42 bootstrap tasks and independently recalculated all 2,869 reported interval rows from hashed NPZ draws.

2.2. Primary precision families across stages and budgets

Table S2 reports all 23 primary policy/block families. The range column is descriptive; the final two columns count cells whose Bonferroni-percentile family-wise interval lies entirely below or above zero. Exact C14B policy and C14D harmonized presentations may refer to the same saved pair and are retained for traceability rather than counted as independent evidence.

Table S2. Primary restricted-minus-full precision families across all frozen stages and budgets. Values in the range are percentage points.

Source	Dataset	Outcome	Policy/block	Cells	Range	Negative	Positive
C14B	OULAD	Fail/withdraw	minimized	24	[-4.06, +1.75]	3	0
C14B	OULAD	Fail/withdraw	gender/disability	24	[-0.54, +0.54]	0	0
C14B	UCI 320 Math.	G3 < 10	no demographic/family	12	[-19.05, +10.08]	0	0
C14B	UCI 320 Math.	G3 < 10	no sensitive/wellbeing	12	[-4.76, +5.04]	0	0
C14B	UCI 320 Math.	G3 < 10	school observable	12	[-14.29, +10.08]	0	0
C14B	UCI 320 Math.	G3 < 10	no sex	12	[-5.00, +5.00]	0	0
C14B	UCI 320 Port.	G3 < 10	no demographic/family	12	[-3.03, +14.71]	0	0
C14B	UCI 320 Port.	G3 < 10	no sensitive/wellbeing	12	[-3.03, +5.88]	0	0
C14B	UCI 320 Port.	G3 < 10	school observable	12	[-3.03, +5.88]	0	0
C14B	UCI 320 Port.	G3 < 10	no sex	12	[-4.55, +2.94]	0	0
C14B	UCI 697	Dropout vs gradu- ate	legacy operational	8	[-34.92, -1.35]	6	0
C14B	UCI 697	Dropout vs gradu- ate	no direct personal	8	[-1.06, +0.00]	0	0
C14B	UCI 697	Dropout vs gradu- ate	no family/financial	8	[-23.81, -1.35]	6	0
C14B	UCI 697	Dropout vs gradu- ate	no gender/special needs	8	[-0.91, +0.81]	0	0
C14D	OULAD	Fail/withdraw	sex/gender	24	[-0.23, +1.54]	0	0
C14D	OULAD	Fail/withdraw	socioeconomic/family	24	[-4.05, +0.94]	2	0
C14D	UCI 320 Math.	G3 < 10	sex/gender	12	[-5.00, +5.00]	0	0
C14D	UCI 320 Math.	G3 < 10	socioeconomic/family	12	[-10.00, +5.04]	0	0
C14D	UCI 320 Port.	G3 < 10	sex/gender	12	[-4.55, +2.94]	0	0
C14D	UCI 320 Port.	G3 < 10	socioeconomic/family	12	[-4.55, +8.82]	0	0
C14D	UCI 697	Dropout vs all other	socioeconomic/family	8	[-21.78, -0.87]	7	0
C14D	UCI 697	Dropout vs gradu- ate	sex/gender	8	[-1.09, +0.81]	0	0
C14D	UCI 697	Dropout vs gradu- ate	socioeconomic/family	8	[-23.81, -1.35]	6	0

2.3. Harmonized block contrasts at the 10% operating point

Table S3 lists every C14D pair at the historically emphasized 10% budget. Pointwise and family-wise intervals use paired multiplicities. “Excludes zero” is a statistical description, not an equivalence or non-inferiority decision.

Table S3. Harmonized restricted-minus-full precision at a 10% review budget. Estimates and intervals are percentage points.

Dataset	Outcome	Stage	Block	Estimate	Pointwise 95%	Family-wise 95%	Excludes zero
OULAD	Fail/withdraw	day14	sex/gender	+0.00	[-1.17, +0.96]	[-1.79, +1.49]	No
OULAD	Fail/withdraw	day28	sex/gender	-0.22	[-1.08, +1.41]	[-1.73, +2.05]	No
OULAD	Fail/withdraw	day42	sex/gender	+0.55	[-0.33, +1.33]	[-0.66, +1.87]	No
OULAD	Fail/withdraw	day56	sex/gender	+0.56	[-0.34, +1.46]	[-0.90, +2.00]	No
OULAD	Fail/withdraw	day70	sex/gender	-0.23	[-0.80, +0.58]	[-1.26, +1.03]	No
OULAD	Fail/withdraw	day90	sex/gender	+0.00	[-0.70, +0.35]	[-1.05, +0.79]	No
OULAD	Fail/withdraw	day14	socioeconomic/family	-3.74	[-5.88, -1.28]	[-7.13, +0.11]	No
OULAD	Fail/withdraw	day28	socioeconomic/family	+0.43	[-1.30, +2.27]	[-2.27, +3.25]	No
OULAD	Fail/withdraw	day42	socioeconomic/family	-0.88	[-2.98, +0.99]	[-4.06, +1.99]	No
OULAD	Fail/withdraw	day56	socioeconomic/family	-0.56	[-1.69, +1.35]	[-2.59, +2.14]	No
OULAD	Fail/withdraw	day70	socioeconomic/family	-0.46	[-1.49, +0.57]	[-2.17, +1.15]	No
OULAD	Fail/withdraw	day90	socioeconomic/family	+0.94	[-0.58, +1.64]	[-1.29, +2.31]	No
UCI 320 Math.	G3 < 10	baseline	sex/gender	-5.00	[-15.00, +7.50]	[-21.95, +12.50]	No
UCI 320 Math.	G3 < 10	period1	sex/gender	+5.00	[-2.44, +10.00]	[-5.00, +12.50]	No
UCI 320 Math.	G3 < 10	period2	sex/gender	+2.50	[-0.00, +7.32]	[-0.00, +10.00]	No
UCI 320 Math.	G3 < 10	baseline	socioeconomic/family	-10.00	[-20.00, +5.00]	[-26.07, +12.50]	No
UCI 320 Math.	G3 < 10	period1	socioeconomic/family	+0.00	[-7.50, +7.50]	[-12.50, +10.00]	No
UCI 320 Math.	G3 < 10	period2	socioeconomic/family	+0.00	[-7.50, +7.32]	[-12.20, +10.00]	No
UCI 320 Port.	G3 < 10	baseline	sex/gender	-3.03	[-9.09, +6.06]	[-13.64, +9.23]	No
UCI 320 Port.	G3 < 10	period1	sex/gender	-4.55	[-9.09, +4.55]	[-10.77, +7.58]	No
UCI 320 Port.	G3 < 10	period2	sex/gender	-1.52	[-3.08, +3.08]	[-4.62, +6.06]	No
UCI 320 Port.	G3 < 10	baseline	socioeconomic/family	-3.03	[-12.31, +7.58]	[-17.66, +12.12]	No
UCI 320 Port.	G3 < 10	period1	socioeconomic/family	-4.55	[-9.09, +4.62]	[-12.12, +9.17]	No
UCI 320 Port.	G3 < 10	period2	socioeconomic/family	-4.55	[-7.58, +3.03]	[-9.23, +4.62]	No
UCI 697	Dropout vs all other	enroll.	socioeconomic/family	-21.78	[-27.17, -17.86]	[-29.09, -15.99]	Yes
UCI 697	Dropout vs all other	semester 1	socioeconomic/family	-5.33	[-7.13, -2.00]	[-8.22, -1.11]	Yes
UCI 697	Dropout vs graduate	enroll.	sex/gender	+0.81	[-1.08, +1.89]	[-1.63, +2.70]	No
UCI 697	Dropout vs graduate	semester 1	sex/gender	+0.00	[-0.81, +0.81]	[-1.08, +1.35]	No
UCI 697	Dropout vs graduate	enroll.	socioeconomic/family	-21.83	[-26.88, -17.07]	[-28.74, -15.10]	Yes
UCI 697	Dropout vs graduate	semester 1	socioeconomic/family	-1.35	[-3.22, +0.27]	[-4.05, +0.81]	No

2.4. Complete primary proxy-probe metrics

Table S4 reports the primary fixed logistic probes in their full held-out cohorts. Average-precision lift divides AP by target prevalence. The UCI 697 educational special-needs target has only 40 positives; its estimates are retained but treated as unstable. OULAD unseen-student and HGB sensitivity rows remain in the machine-readable C14C results.

Table S4. Primary fixed logistic proxy probes.

Dataset	Stage	Target	n	Positive	Prev.	AUROC	AP	AP lift	Bal. acc.
OULAD	day14	disability	9331	868	0.093	0.680	0.156	1.68	0.641
OULAD	day28	disability	9211	854	0.093	0.667	0.146	1.57	0.630
OULAD	day42	disability	9040	833	0.092	0.667	0.142	1.54	0.615
OULAD	day56	disability	8877	811	0.091	0.671	0.148	1.62	0.626
OULAD	day70	disability	8694	783	0.090	0.663	0.137	1.52	0.626
OULAD	day90	disability	8532	761	0.089	0.678	0.148	1.66	0.631
OULAD	day14	gender	9331	5425	0.581	0.800	0.818	1.41	0.743
OULAD	day28	gender	9211	5339	0.580	0.690	0.770	1.33	0.666
OULAD	day42	gender	9040	5215	0.577	0.766	0.757	1.31	0.747
OULAD	day56	gender	8877	5127	0.578	0.811	0.819	1.42	0.750
OULAD	day70	gender	8694	5010	0.576	0.791	0.808	1.40	0.738
OULAD	day90	gender	8532	4905	0.575	0.812	0.823	1.43	0.748
UCI 320 Math.	baseline	sex	395	187	0.473	0.746	0.709	1.50	0.670
UCI 320 Math.	period1	sex	395	187	0.473	0.750	0.712	1.50	0.665
UCI 320 Math.	period2	sex	395	187	0.473	0.748	0.710	1.50	0.658
UCI 320 Port.	baseline	sex	649	266	0.410	0.734	0.667	1.63	0.643
UCI 320 Port.	period1	sex	649	266	0.410	0.734	0.673	1.64	0.655
UCI 320 Port.	period2	sex	649	266	0.410	0.734	0.673	1.64	0.664
UCI 697	enroll.	Educational needs	special 3630	40	0.011	0.637	0.020	1.77	0.591
UCI 697	semester 1	Educational needs	special 3630	40	0.011	0.655	0.021	1.88	0.575
UCI 697	enroll.	Gender	3630	1249	0.344	0.771	0.650	1.89	0.695
UCI 697	semester 1	Gender	3630	1249	0.344	0.774	0.654	1.90	0.701

2.5. C14 interpretation guards

The following guards are part of the frozen reporting contract: residual predictability is not actual proxy use; operational harmonization is not measurement equivalence; a family-wise interval crossing zero is not equivalence; exact policy/block duplicates are

not independent confirmations; and no result is a fairness certification, privacy guarantee, formal non-inferiority conclusion, or causal estimate of teacher or student benefit. OULAD, UCI 697, and UCI 320 remain separately identified throughout.

2.6. Predictor inventory

Context predictors.

Module, number of previous attempts, studied credits, and registration lead days.

Click/activity and assessment-process predictors.

Total clicks; active days; mean, standard deviation, and maximum daily clicks; first and last activity day; clicks and active days in the last seven days and the preceding seven days; pre-start clicks and active days; inactivity days; log click trend; assessments submitted; assessment weight completed; late submissions; assessments expected; assessment weight expected; assessment submission ratio; and assessment weight-completion ratio. Structurally all-missing training columns at a landmark were omitted.

Full-policy background fields.

Gender, region, highest education, Index of Multiple Deprivation band, age band, and disability.

Strict original tier.

Module, previous attempts, studied credits, registration lead days, assessments submitted, assessment weight completed, late submissions, assessments expected, assessment weight expected, assessment submission ratio, and assessment weight-completion ratio.

Strict enriched tier.

The enriched policy retained click-independent context and added only typed process features whose due date and submission availability were no later than the landmark. It excluded scores, click/activity telemetry, and future records. A separately typed alternative and an expanded leaf grid were retained as unsuccessful sensitivities.

UCI 697 policies.

The full enrollment set comprised 24 fields. The legacy operational set retained six admissions/course fields. Direct-personal exclusion removed marital status, nationality, displacement, educational special-needs status, gender, age at enrollment, and international status. Family/financial exclusion removed parental qualifications and occupations, debtor and tuition status, and scholarship status. Targeted exclusion removed gender and special-needs status. Semester-one policies added the six first-semester curricular-unit fields; second-semester fields were prohibited.

UCI 320 policies.

Full policies used all non-target fields available at baseline, with G1, or with G1+G2. Demographic/family exclusion removed sex, age, address, family size and parental status, parental education and occupation, and guardian. Sensitive/wellbeing exclusion removed romantic relationship, family relationship, weekday and weekend alcohol use, and health. The school-observable tier retained school, reason, study time, prior failures, school support, paid classes, activities, absences, and grades available at the stage. These are frozen operational group names rather than claims of intrinsic sensitivity or legality.

2.7. Official-data main-policy point estimates

77

Table S5. True alerts, alert counts, and Precision@10% for the three main policies.

Day	Full TP/n	Full P	Six-field TP/n	Six-field P	G/D-excl. TP/n	G/D-excl. P
14	708/935	0.7572	670/935	0.7166	709/935	0.7583
28	764/923	0.8277	762/923	0.8256	763/923	0.8267
42	769/908	0.8469	762/908	0.8392	765/908	0.8425
56	801/890	0.9000	804/890	0.9034	803/890	0.9022
70	815/873	0.9336	811/873	0.9290	813/873	0.9313
90	801/855	0.9368	809/855	0.9462	804/855	0.9404

2.8. UCI 697 workload-policy contrasts

78

Table S6. UCI 697 HGB restricted-minus-full Precision@10% differences for the primary outcome. Values are percentage points.

Stage	Policy	Difference	Pointwise 95%	Bonferroni 95%
Enrollment	Legacy operational	-32.61	[-38.44, -27.76]	[-39.25, -27.06]
Enrollment	No direct personal	-0.54	[-2.71, 1.34]	[-3.11, 1.48]
Enrollment	No family/financial	-21.83	[-26.95, -17.30]	[-27.57, -16.67]
Enrollment	No gender/special needs	+0.81	[-1.08, 2.16]	[-1.35, 2.42]
Semester 1	Legacy operational	-1.35	[-3.76, 0.00]	[-4.04, 0.00]
Semester 1	No direct personal	+0.00	[-1.08, 0.82]	[-1.08, 1.08]
Semester 1	No family/financial	-1.35	[-3.23, 0.27]	[-3.50, 0.54]
Semester 1	No gender/special needs	+0.00	[-0.81, 0.81]	[-0.81, 1.08]

2.9. UCI 320 workload-policy contrasts

79

Table S7. UCI 320 HGB restricted-minus-full Precision@10% differences. Subjects are separate; values are percentage points.

Subject	Stage	Policy	Difference	Bonferroni 95%
Mathematics	Baseline	No demographic/family	-7.50	[-25.00, 12.50]
Mathematics	Baseline	No sensitive/wellbeing	+0.00	[-12.50, 17.50]
Mathematics	Baseline	School-observable only	+0.00	[-20.00, 25.00]
Mathematics	Baseline	No sex	-5.00	[-17.50, 10.00]
Mathematics	+G1	No demographic/family	-2.50	[-12.20, 7.50]
Mathematics	+G1	No sensitive/wellbeing	+5.00	[-7.50, 12.50]
Mathematics	+G1	School-observable only	-5.00	[-17.50, 7.50]
Mathematics	+G1	No sex	+5.00	[-2.50, 12.50]
Mathematics	+G1+G2	No demographic/family	+0.00	[-7.50, 7.50]
Mathematics	+G1+G2	No sensitive/wellbeing	-2.50	[-9.76, 2.44]
Mathematics	+G1+G2	School-observable only	+0.00	[-7.50, 7.50]
Mathematics	+G1+G2	No sex	+2.50	[0.00, 7.50]
Portuguese	Baseline	No demographic/family	+0.00	[-9.23, 13.64]
Portuguese	Baseline	No sensitive/wellbeing	-3.03	[-15.15, 9.23]
Portuguese	Baseline	School-observable only	+0.00	[-12.18, 13.71]
Portuguese	Baseline	No sex	-3.03	[-10.77, 7.69]
Portuguese	+G1	No demographic/family	-3.03	[-7.58, 8.17]
Portuguese	+G1	No sensitive/wellbeing	+1.52	[-6.06, 13.64]
Portuguese	+G1	School-observable only	-3.03	[-12.12, 9.14]
Portuguese	+G1	No sex	-4.55	[-10.61, 4.62]
Portuguese	+G1+G2	No demographic/family	-1.52	[-7.58, 4.62]
Portuguese	+G1+G2	No sensitive/wellbeing	+0.00	[-6.15, 7.58]

Subject	Stage	Policy	Difference	Bonferroni 95%
Portuguese	+G1+G2	School-observable only	+1.52	[-6.06, 9.09]
Portuguese	+G1+G2	No sex	-1.52	[-4.55, 4.55]

All UCI 320 workload intervals in Table S7 include or touch zero. The isolated family-wise HGB finding discussed in the main text concerns AUROC, not workload precision. The complete machine-readable supplement retains AP, AUROC, Brier, precision, recall, pointwise intervals, familywise intervals, and all bootstrap differences.

2.10. Leave-one-field-out sensitivity

Table S8. Leave-one-field-out minus full Precision@10% contrasts.

Day	Excluded field	Difference	Pointwise 95% interval	Bonferroni lower bound
14	Age band	+0.0011	[-0.0096, 0.0107]	-0.0117
14	Disability	+0.0021	[-0.0117, 0.0117]	-0.0167
14	Gender	+0.0000	[-0.0117, 0.0096]	-0.0149
14	Highest education	-0.0342	[-0.0544, -0.0085]	-0.0619
14	IMD band	-0.0043	[-0.0204, 0.0128]	-0.0264
14	Region	-0.0043	[-0.0128, 0.0106]	-0.0168
42	Age band	+0.0077	[-0.0055, 0.0143]	-0.0079
42	Disability	+0.0011	[-0.0099, 0.0111]	-0.0143
42	Gender	+0.0055	[-0.0033, 0.0143]	-0.0055
42	Highest education	-0.0066	[-0.0220, 0.0132]	-0.0278
42	IMD band	+0.0011	[-0.0144, 0.0187]	-0.0198
42	Region	-0.0033	[-0.0143, 0.0078]	-0.0177

2.11. Common-cohort timing sensitivity

The common4 cohort retained registrations present at days 14, 28, 42, and 56. The common6 cohort retained registrations present at all six horizons. Each row in Table S9 is a day-42-minus-day-14 difference-in-differences in Precision@10%. Positive values indicate that the candidate-minus-full gap became less negative or more positive by day 42.

Table S9. Common-cohort day-42-minus-day-14 difference-in-differences.

Cohort	Candidate	Estimate	Pointwise 95% interval	Bonferroni 95% interval
common4	Six-field excluded	+0.0416	[0.0022, 0.0676]	[-0.0000, 0.0742]
common4	Gender/disability excluded	+0.0000	[-0.0169, 0.0180]	[-0.0180, 0.0202]
common6	Six-field excluded	+0.0281	[0.0011, 0.0622]	[-0.0035, 0.0646]
common6	Gender/disability excluded	+0.0070	[-0.0129, 0.0234]	[-0.0175, 0.0257]

The pointwise complete-exclusion estimates were positive, but the family sensitivity lower bounds were -0.000014 and -0.003517 . The gender/disability-exclusion intervals included zero. These findings do not establish a monotonic timing law.

2.12. Alert-list overlap

Table S10. Candidate versus full alert-list identity. Each replacement count is per direction.

Day	Candidate	Alerts/list	Intersection	Jaccard	Replacements
14	Six-field excluded	935	645	0.5265	290
14	Gender/disability excluded	935	881	0.8908	54
28	Six-field excluded	923	764	0.7061	159
28	Gender/disability excluded	923	862	0.8760	61
42	Six-field excluded	908	739	0.6862	169
42	Gender/disability excluded	908	871	0.9217	37
56	Six-field excluded	890	781	0.7818	109
56	Gender/disability excluded	890	855	0.9243	35
70	Six-field excluded	873	807	0.8594	66
70	Gender/disability excluded	873	852	0.9530	21
90	Six-field excluded	855	797	0.8729	58
90	Gender/disability excluded	855	838	0.9610	17

2.13. Superseded C11 to official-data C12 transition

Table S11 reports the effect of correcting the raw archive while retaining the C11/C12 production code. These are not same-run model contrasts and are not used to estimate a policy effect.

Table S11. True-alert transition from the superseded derivative-input run to C12.

Day	Policy	C11 TP	C12 TP	TP change	Precision change (points)
14	full	705	708	+3	+0.32
14	minimized	670	670	+0	+0.00
14	partial_gender_disability	702	709	+7	+0.75
28	full	773	764	-9	-0.98
28	minimized	762	762	+0	+0.00
28	partial_gender_disability	762	763	+1	+0.11
42	full	770	769	-1	-0.11
42	minimized	762	762	+0	+0.00
42	partial_gender_disability	765	765	+0	+0.00
56	full	800	801	+1	+0.11
56	minimized	804	804	+0	+0.00
56	partial_gender_disability	803	803	+0	+0.00
70	full	816	815	-1	-0.11
70	minimized	811	811	+0	+0.00
70	partial_gender_disability	815	813	-2	-0.23
90	full	805	801	-4	-0.47
90	minimized	809	809	+0	+0.00
90	partial_gender_disability	805	804	-1	-0.12

2.14. Historical numerical replay retained as a failed check

Inherited C06/C08 values cannot be verified without the original complete package. Table S12 is therefore a failed numerical replay record, not a target for tuning and not evidence combined with C12.

Table S12. Inherited NOT_REVERIFIED true-alert counts versus C12.

Day	Policy	Historical TP	C12 TP	Difference	Exact match
14	full	705	708	+3	No
14	minimized	670	670	+0	Yes
14	partial_gender_disability	702	709	+7	No
28	full	764	764	+0	Yes

Day	Policy	Historical TP	C12 TP	Difference	Exact match
28	minimized	761	762	+1	No
28	partial_gender_disability	764	763	-1	No
42	full	770	769	-1	No
42	minimized	762	762	+0	Yes
42	partial_gender_disability	765	765	+0	Yes
56	full	800	801	+1	No
56	minimized	798	804	+6	No
56	partial_gender_disability	804	803	-1	No
70	full	816	815	-1	No
70	minimized	811	811	+0	Yes
70	partial_gender_disability	815	813	-2	No
90	full	805	801	-4	No
90	minimized	810	809	-1	No
90	partial_gender_disability	805	804	-1	No
42	strict_original	671	669	-2	No
42	strict_enriched_due_process	712	674	-38	No

2.15. Reproducibility inventory

102

Table S13. C12–C14 production and verification inventory.

Item	Recorded status
Official OULAD archive	SHA-256 f2ed190 . . . ; ZIP CRC passed; eight members including OULAD .names
Input audit	54/54 checks passed; member hashes and schemas retained
Processed snapshots	Six compressed snapshots plus manifest and hashes
Model search	105 validation candidate fits
Locked artifacts	36 selected pipelines, 36 validation prediction files, and 36 test prediction files
Model/prediction verification	861/861 checks passed before bootstrap
Bootstrap	2,000 dynamic draws per horizon except 5,000 at day 42; 2,000 common4 and 2,000 common6 draws; seed 20260901
Statistical verification	1,954/1,954 checks passed, including draw identity and interval replay
Negative results	Original strict-gate failure, typed alternative, expanded-grid identity, historical replay failure, and global-metric losses retained
Interpretive status	Post-development exploratory; no independent confirmation or formal non-inferiority
C13 inputs	UCI 697 outer SHA-256 e90e55 . . . ; UCI 320 outer SHA-256 82ae9d . . . ; CRC and member hashes passed
C13 locked artifacts	114 model bundles and 243,220 OOF rows saved before statistics; 114/114 full prediction replay passed
C13 paired statistics	470 contrasts, 284,000 bootstrap differences, and 195 attenuation rows; independent verification passed
C13 negative results	First portability serialization FAIL retained; UCI 320 imprecision, null contrasts, and isolated favorable alternatives retained
C14B budget audit	150 configurations reconstructed from saved probabilities; 600 budget metric rows and 484 restricted-minus-full budget contrasts; no fits
C14C proxy audit	44 fixed probe configurations, 250,044 held-out predictions, and 68 metric rows; independent replay error 1.11×10^{-16}
C14D harmonized models	28 new HGB bundles, 408 candidate fits, 12 full controls, and 30 paired block comparisons; no UCI 697 or logistic refits
C14E paired statistics	151 contrasts in 42 shared-stream tasks; 2,869 interval rows independently recalculated from hashed draws
C14 primary precision	296 stage–budget rows; 30 family-wise negative and zero family-wise positive cells, including planned exact duplicate views
C14 interpretation	Post-development; residual predictability is not actual proxy use; harmonization is not measurement equivalence; no equivalence, fairness, privacy, or causal claim